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Hemodialysis: Principles and Techniques

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Hemodialysis (HD) has been in practice for more than 50 years and has extended the lives of millions of patients with kidney failure worldwide. Although the basic principles of HD are still being applied today, dialysis technology has improved markedly. The main components of the dialysis system are the extracorporeal blood circuit, the dialyzer, the dialysis machine, and the water purification system.

DIALYZER DESIGNS AND MEMBRANES

The dialyzer provides countercurrent transfer of solutes and fluid across a semipermeable membrane. The semipermeable dialysis membrane separates the blood compartment from the dialysate compartment (Table 98.1). Most current membranes are made of entirely synthetic materials, such as polyacrylonitrile, polysulfone, polycarbonate, polyamide, and polymethylmethacrylate, and are more biocompatible than cellulose-based membranes.

Transport of molecules across the dialysis membrane is driven by (1) the concentration gradient (diffusive transport) and (2) the hydrostatic pressure gradient across the membrane (convective transport). In the case of protein-bound solutes, only their free fraction is transported across the membrane, unless so-called albumin-leaky membranes are used. The flux of solutes is also affected by their charge and the protein concentration on the blood side (Gibbs-Donnan effect). Dialyzer efficiency increases with surface area (usually between 0.8–2.1 m^2; smaller dialyzers with surface areas between 0.075–0.245 m^2 are used for treatment of neonates). The dialyzer mass transfer area coefficient (KoA) for urea is a measure of the theoretically maximal possible urea clearance (mL/min). Dialyzer efficiency can be categorized based on KoA for urea as low (<500 mL/min), moderate (500–700 mL/min), and high (>700 mL/min). High-flux and medium cut-off membranes have pores large enough to allow some passage of molecules such as β_2-microglobulin (molecular weight 11,800 Da), tumor necrosis factor (TNF)-α (17 kDa), and beyond.

Water permeability is defined by the ultrafiltration coefficient (Kuf) that describes the transmembrane ultrafiltration volume per hour and unit of hydrostatic transmembrane pressure in millimeters of mercury. In high-flux dialyzers, Kuf can be as high as 80 mL/h/mm Hg. High-flux and medium cut-off membranes demonstrate substantial internal ultrafiltration at the proximal part of the dialyzer and backfiltration of dialysate into the blood at the distal end of the dialyzer blood compartment. Therefore, water quality is of paramount importance when high-flux and medium cut-off dialyzers are used (see later).

SAFETY MONITORS

Safety monitors are integral parts of the dialysis machine.

Pressure monitors measure hydrostatic pressures in critical positions (Fig. 98.1)[1]:

- Between the arterial needle and the blood pump (prepump pressure): Overly negative values may signal reduced arterial inflow and access problems.
- Between the blood pump and the dialyzer inlet (postpump): A high pressure may signal dialyzer clotting.
- Between the dialyzer outlet and the air trap (venous pressure): A high pressure may point toward an obstruction in the venous limb; it is important to recognize that in the event of venous needle displacement, with external bleeding, the venous pressure will remain positive.

Venous air detectors and *air trap* are located downstream of the venous pressure monitor. A positive signal at the air detector automatically clamps the venous line and stops the blood pump.

A *blood leak detector* is placed in the dialysate outflow line. Dialysate temperature is constantly monitored. Dialysate is produced by a proportioning system that mixes acid and bicarbonate concentrates with highly purified water. The concentration of ions in the dialysate translates into conductivity, which is measured by a *dialysis conductivity monitor*.

DIALYSATE FLUID

Water and Water Treatment

A standard 4-hour HD session exposes the patient to 120 to 160 L of water at a rate of 600 to 800 mL/min. A typical water purification plant is shown in Fig. 98.2. Water from municipal sources is filtered to remove particulate matter. Activated carbon devices, which need regular replacement, adsorb substances such as endotoxins, chlorine, and chloramines. Downstream water softeners use a resin coated with sodium ions, which are exchanged for calcium and magnesium ions before the water enters the reverse osmosis (RO) system. During RO, water is pumped at high pressure (15–20 bar) through a membrane with pores of less than 1.0 nm diameter, providing an absolute barrier for molecules larger than 100 to 300 Da. Depending on the technology used, 20% to 60% of water is rejected by the RO, meaning that water requirements are much higher than the final dialysate volume.

Standards for chemical quality of water are widely accepted, but there is less consensus regarding acceptable levels of bacterial and endotoxin contamination. The microbiologic standards for HD water, dialysis fluid, and substitution fluid vary across countries (Table 98.2).[2] Municipal water supplies may contain undesirable added substances such as aluminum and chloramines. Aluminum accumulation may result in neurologic disorder, bone disease, and erythropoietin-resistant anemia. Plasma aluminum levels should be less than 1 μmol/L. Chloramines have been associated with hemolysis and methemoglobinemia. Copper and zinc may leach from plumbing components and may cause hemolysis. Lead has been associated with abdominal pain and muscle weakness. Nitrate and nitrite may cause nausea and

TABLE 98.1 Dialysis Membrane Properties

Membrane	Membrane Name (Example)	High or Low Flux	Biocompatibility
Cellulose	Cuprophan	Low	Low
Semisynthetic cellulose			
Cellulose diacetate	Cellulose acetate	High and low	Intermediate
Cellulose triacetate	Cellulose triacetate	High	Good
Diethylaminoethyl-substituted cellulose	Hemophane	High	Intermediate
Synthetic polymers			
Polymethylmethacrylate	PMMA	High	Good
Polyacrylonitrile methacrylate copolymer	PAN	High	Good
Polyacrylonitrile methallyl sulfonate copolymer	PAN/AN-69	High	Good
Polyamide	Polyflux	High and low	Good
Polycarbonate-polyether	Gambrane	High	Good
Ethylene vinyl alcohol copolymer	EVAL	High	Good
Polysulfone	Polysulfone	High and low	Good

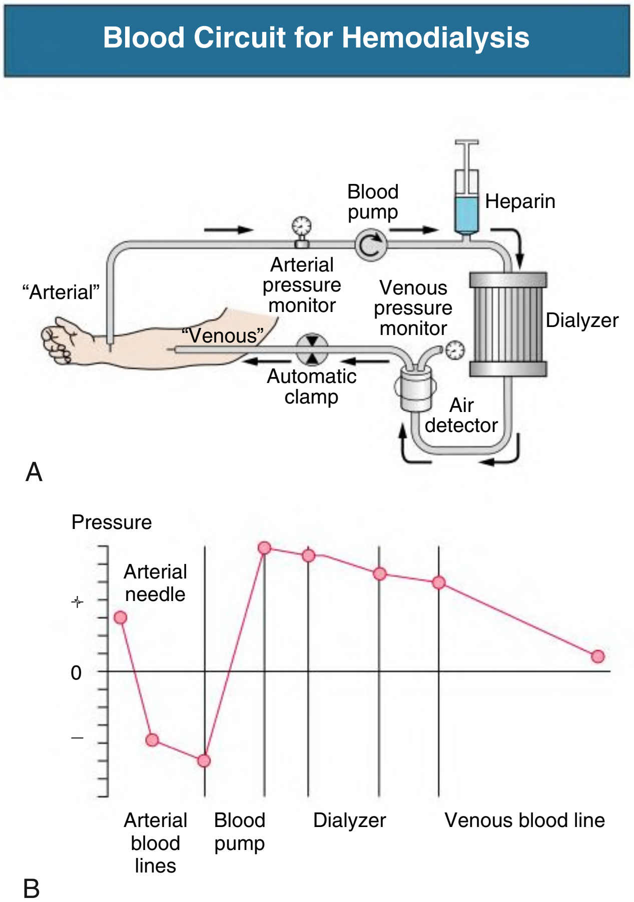

Fig. 98.1 Blood Circuit for Hemodialysis. (A) The blood circuit. (B) The pressure profile in the blood circuit with an arteriovenous fistula as the vascular access.

seizures. High concentrations of calcium may cause the so-called *hard water syndrome,* a condition characterized by nausea, vomiting, blood pressure instability, and fatigue during and after dialysis.

Gram-negative bacteria produce endotoxins, and fragments of these endotoxins may be responsible for some dialysis-related symptoms. Exposure to endotoxin is associated with rigors, hypotension, and fever. Use of a polysulfone or polyamide filter in the dialysate line may be adequate to remove endotoxins. Liquid bicarbonate dialysis fluid concentrate distributed in a central distribution system may be a source of bacterial growth and should be replaced daily; acid concentrates in canisters and bicarbonate powder represent no bacterial growth risk.

Ultrapure water is recommended for use with high-flux dialyzers. Water used for online hemodiafiltration (HDF) must be virtually sterile and nonpyrogenic, thus fulfilling major criteria for ultrapure water.

Dialysate Solution

Dialysate is made by mixing two components, which may be provided as liquid or dry (powder) concentrates (Table 98.3). The *base concentrate* contains sodium bicarbonate and sodium chloride. The *acid concentrate* typically contains chloride salts of sodium, calcium, magnesium, and potassium, glucose monohydrate, and an organic acid, the latter in the form of acetic acid, citric acid, or lactic acid. The acid concentrate may contain the salt of an organic acid, such as sodium acetate. The purpose of the acid is to lower the dialysate pH to less than 7.3 so that calcium and magnesium do not precipitate when bicarbonate is added. Base and acid components are mixed simultaneously with purified water to make the dialysate. Dialysate proportioning pumps ensure proper mixing. One typical mixing relationship is 1:1.72:42.28 (acid concentrate to base concentrate to water; the so-called *45X preparation*). Some dry acid concentrates contain sodium diacetate (8 mmol/L), which is composed of equal parts of acetic acid and sodium acetate. After mixing with bicarbonate, the final dialysate contains 8 mmol/L sodium acetate.[3] Dialysate composition can be modified by small changes in the mixing ratio and adding salt solutions; potential advantages and disadvantages of dialysate modifications are shown in Table 98.4.

BIOCOMPATIBILITY

The contact of blood with some lines and membranes triggers an inflammatory response. Although many components of the dialysis procedure contribute to the degree of biocompatibility, it is the membrane itself that is most important. Biocompatibility is especially important when cellulose membranes are used, whereas synthetic and reused membranes activate complement to a much lesser extent (Fig. 98.3). Activation of complement peaks at 15 minutes after the start of dialysis and lasts up to 90 minutes. Symptoms associated with activation of complement system and proinflammatory pathways include shortness of breath, chest pain, headache, nausea, vomiting, and hypotension.

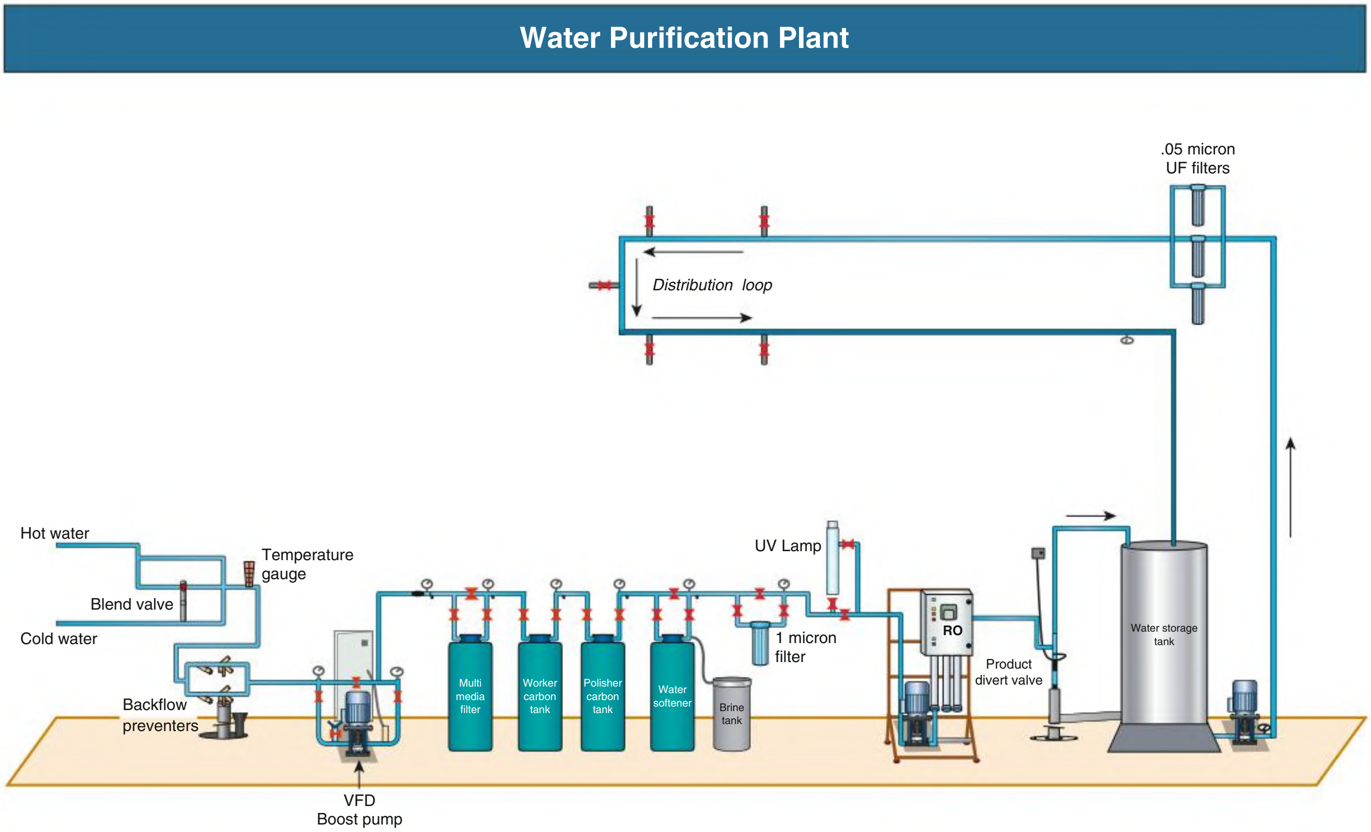

Fig. 98.2 **Water Purification Plant.** The four main functional domains are water supply, water pretreatment, primary purification, and delivery to the loop. A series of filters and water softeners are the key technical pretreatment components, whereas reverse osmosis is the main component of the primary purification. Note that several features are optional, such as the water storage tank. *UF,* Ultrafiltration; *UV,* ultraviolet light; *VFD,* variable frequency drive. (Courtesy Rob Levin and Randy Hux, New York, NY.)

TABLE 98.2 Microbiologic Standards for Water, Concentrates, and Dialysis Fluids

National and International Standards	Year Issued	Microorganisms (CFU/mL)	Endotoxins (EU/mL)
Water			
EDTA-ERA	2001	<100	<0.25
USA (AAMI RD 52) (minimum regulatory requirement)	2004	<200 (action level≥ 50)	<2 (action level ≥ 1)
ISO/DIS 13959 (preferred recommendation)	2014	<100 (action level t≥ 50)	<0.25 (action level ≥ 0.125)
Concentrates (Acid and Basic)			
USA (AAMI RD 52)	2004	<200 (action level≥ 50)	2 (action level ≥ 1)
European Pharmacopoeia, 5th ed.	2005	—	<0.5[a]
Standard Dialysis Fluid			
EDTA-ERA	2001	<100	<0.25
USA (AAMI RD 52)	2004	<100 (action level≥ 50)	<0.5 (action level ≥ 0.25)
ISO/FDIS 23500-5:2018(E)	2018	<100 (action level≥ 50)	<0.5 (action level ≥ 0.25)
Ultrapure Dialysis Fluid Before Last Filter for Hemodiafiltration Online			
EDTA-ERA	2001	<0.1	<0.03
USA (AAMI RD 52)	2004	<0.1	<0.03
ISO/FDIS 23500-5:2018(E)	2018	<0.1	<0.03
Substitution Fluid Online			
EDTA-ERA	2001	$<10^{-6}$	<0.25
USA (AAMI RD 52)	2004	$<10^{-6}$	<0.03
ISO/FDIS 23500-5:2018(E)	2018	Sterile	Nonpyrogenic

[a]Diluted to user concentration.

AAMI, Association for the Advancement of Medical Instrumentation; *EDTA-ERA,* European Dialysis and Transplant Association European Renal Association; *ISO/FDIS,* International Organization for Standardization/Final Draft International Standard; *RD,* renal disease.

HEMOFILTRATION, HEMODIAFILTRATION, AND INTERNAL FILTRATION AND BACKFILTRATION

Hemofiltration (HF) and HDF involve the removal of large fluid volumes from the patient, with the removed fluid replaced by substitution fluid. It is the use of substitution fluid that sets these techniques apart from simple ultrafiltration.

TABLE 98.3 Composition of Dialysates for Bicarbonate Dialysis

	CONCENTRATION	
Component	**Range**	**Typical**
Electrolytes (mmol/L)		
Sodium	135–145	138
Potassium	1.0–4.0	2.0
Calcium	1.0–1.75	1.25
Magnesium	0.5–1.0	0.75
Chloride	87–124	105
Buffers (mmol/L)		
Acetate	2–4	3
Bicarbonate	20–40	35
pH	7.1–7.3	7.2
P_{CO_2} (mm Hg)	40–100	
Glucose	0–11 (0–200 mg/dL)	5.5 (100 mg/dL)

HF (Fig. 98.4) is a convective elimination technique by which water and solutes from the blood compartment are driven solely by positive hydrostatic pressure across the dialyzer membrane into the filtrate compartment without the use of dialysate. As a result of solvent drag effects, small and larger solutes are eliminated at a rate depending on membrane characteristics.

HDF (Fig. 98.5) combines diffusive (HD) and convective (HF) solute transport using a high-flux membrane. Fluid is removed by ultrafiltration (convective volume) and the volume of filtered fluid exceeding the volume to achieve target weight loss is replaced by ultrapure infusion solution (substitution volume). *Online* HDF refers to the online production by the dialysis machine of ultrapure, nonpyrogenic dialysate, which is also used as infusion solution. *High-volume* HDF refers to an effective convection volume of more than 23 L per dialysis. HDF modalities are further characterized by the site of infusion of substitution fluid.

Postdilution Hemodiafiltration

The substitution fluid is infused to the patient downstream of the dialyzer. Filtration is limited by hemoconcentration in the dialyzer. Postdilution HDF is most efficient in terms of increasing solute removal.

Predilution Hemodiafiltration

Substitution fluid is added upstream of the dialyzer and immediately removed by convection. For an identical substitution volume, the efficiency of this mode is lower than that of postdilution HDF because of the dilution of solute blood concentrations before the dialyzer and consecutive and thus a reduction of in the transmembrane gradient.

TABLE 98.4 Advantages and Disadvantages of Modifications in the Dialysate Composition

Component	Advantage	Disadvantage
Sodium		
Increased	Hemodynamic stability	Postdialytic thirst; increased postdialytic serum sodium levels; increased intradialytic weight gain; high blood pressure
Decreased	Reduced osmotic stress in the presence of predialytic hyponatremia	Intradialytic hemodynamic instability
Potassium		
Increased	Fewer arrhythmias in digoxin intoxication with hypokalemia; may improve hemodynamic stability	Hyperkalemia
Decreased	Increased dietary potassium intake	Arrhythmias; risk for sudden death
Calcium		
Increased	Suppresses PTH, increased hemodynamic stability	Hypercalcemia, vascular calcification, adynamic bone disease resulting from PTH suppression
Decreased	Permits more liberal use of calcium-containing phosphate binders	Stimulation of PTH, reduced hemodynamic stability
Bicarbonate		
Increased	Acidosis control improved	Postdialytic alkalosis; increased mortality
Decreased	No postdialytic alkalosis	Promotes acidosis; increased mortality
Magnesium		
Increased	Hemodynamic stability, fewer arrhythmias, suppresses PTH	Altered nerve conduction, pruritus, renal bone disease
Decreased	Permits use of magnesium-containing phosphate binders; improved bone mineralization; less bone pain	Arrhythmias, muscle weakness and cramps, elevated PTH
Glucose		
Decreased	Avoidance of intradialytic hyperglycemia and hyperinsulinemia	Increased risk for disequilibrium (rare), hypoglycemia
Increased	Lower risk for disequilibrium	Intradialytic hyperglycemia and hyperinsulinism
Citrate	Heparin-sparing effect	High blood citrate levels in liver failure

PTH, Parathyroid hormone.

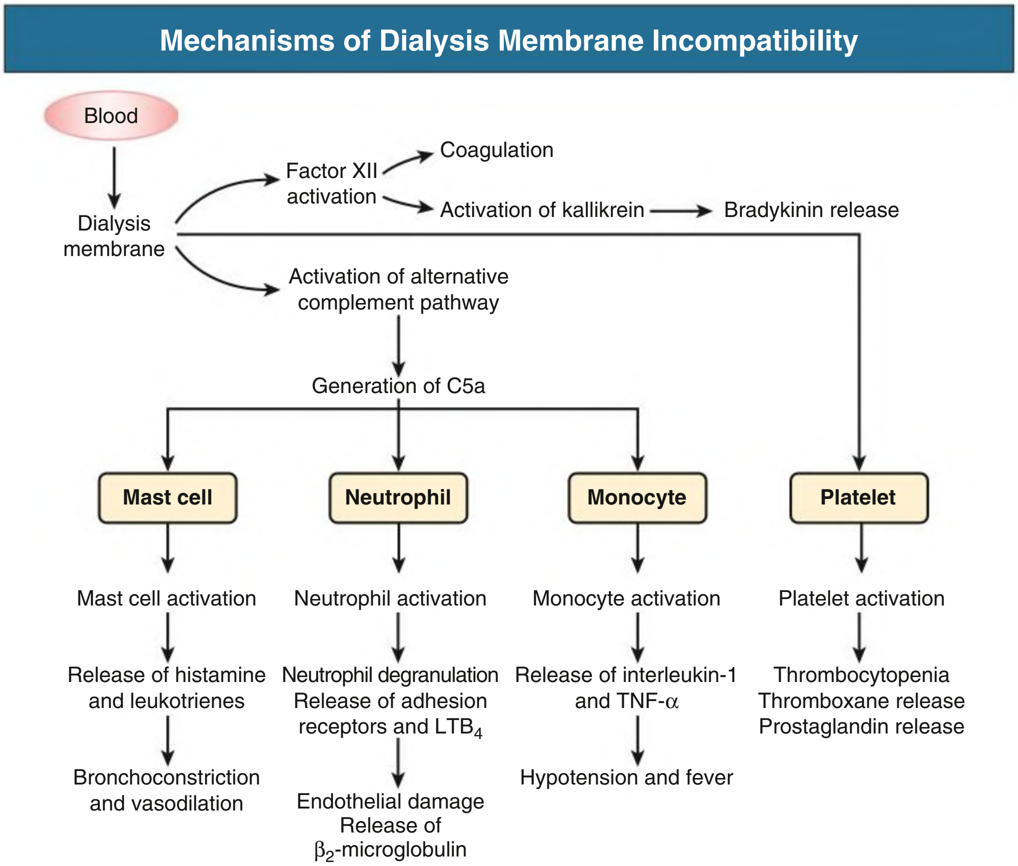

Fig. 98.3 Mechanisms of Dialysis Membrane Incompatibility. Pathways involved in the body's response to dialysis membranes. *LTB_4,* Leukotriene B_4; TNF-α tumor necrosis factor-α.

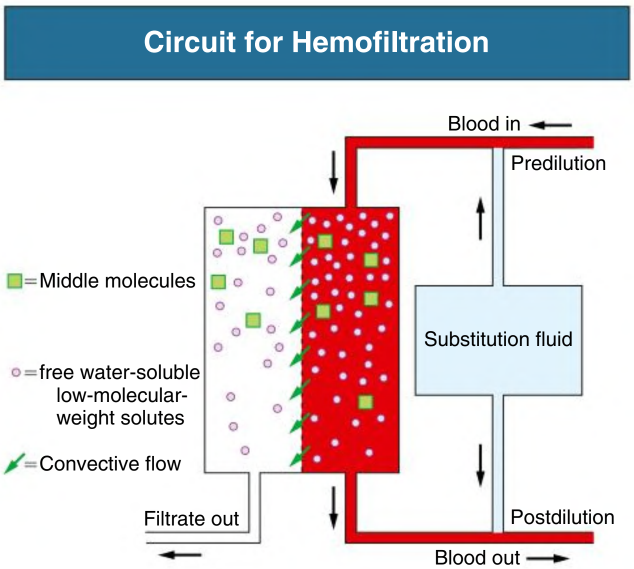

Fig. 98.4 Circuit for Hemofiltration. Substitution fluid is typically administered either predilution or postdilution but not both.

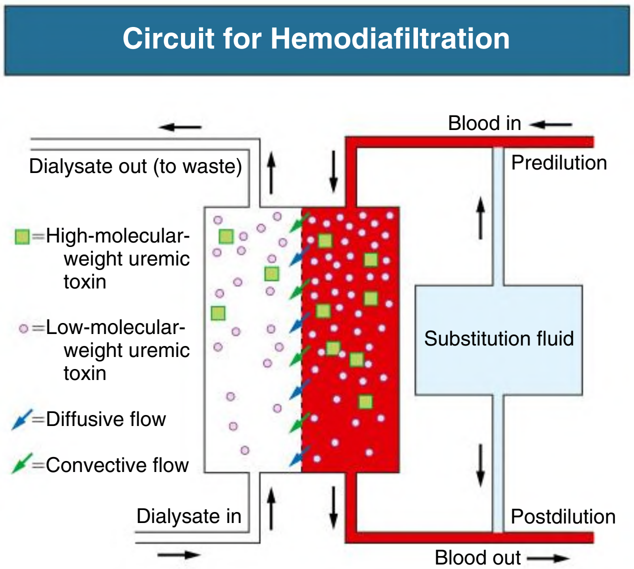

Fig. 98.5 Circuit for Hemodiafiltration. Substitution fluid is typically administered either predilution or postdilution but not both.

To achieve comparable efficiency to postdilution HDF, substitution volume needs to be doubled.

Mixed Dilution Hemodiafiltration

Replacement fluid is infused both upstream and downstream of the dialyzer. The ratio of upstream and downstream infusion rates can be varied to achieve the optimal compromise between maximizing clearance and avoiding the consequences of a high transmembrane pressure and hemoconcentration.

Mid-Dilution Hemodiafiltration

Replacement fluid is infused within specifically designed dialyzers partway down the blood pathway so that the first part of the blood circuit is operated in postdilution and the second part in predilution mode.

Internal Filtration and Backfiltration

Ultrafiltration does not occur in a uniform manner across the entire length of a dialyzer. Several characteristics such as blood and dialysate flows, membrane characteristics, and dialyzer geometries define the internal flow characteristics. In high-flux dialyzers, ultrafiltration dominates in the proximal (upstream) part, whereas backfiltration of dialysate takes place more distally (downstream). The backfiltration has raised concerns in the absence of ultrapure dialysate because endotoxin fragments and other bacterial substances may enter from the dialysate into the patient's blood, possibly aggravating inflammatory response. Medium cut-off membranes were designed for high rates of filtration and backfiltration with the expectation that their use would enhance clearance of middle molecules. A recent randomized controlled trial (RCT) showed larger reduction ratios for complement factor D, free k light chains, TNF-α, and β_2-microglobulin with a medium cut-off membrane versus a high-flux dialyzer.[4] In a cross-over RCT, after 3 months of medium cut-off membrane use, a significant decrease in serum levels of albumin, 25-hydroxyvitamin D and 1,25-dihydroxyvitamin D was seen,[5] possibly related to the loss of vitamin D–binding protein (52 kDa).

UREMIC TOXINS AND THEIR REMOVAL BY HEMODIALYSIS

Uremic toxins have been traditionally categorized based on their molecular weight and binding properties (Table 98.5). Recent research has identified the gut, in particular the colon, as an important source of uremic toxins and their precursors. These colon-derived toxins are products of bacterial metabolisms, such as phenols, indoles, and amines.

Low-molecular-weight compounds not bound to protein: These include urea and guanidines such as asymmetric dimethylarginine (ADMA), purines, pyrimidines, oxalate, phosphorus, and uric acid. Modern dialysis membranes achieve maximal removal of these compounds.

Low-molecular-weight compounds bound to protein: These include indoxyl and cresyl compounds, advanced glycation

TABLE 98.5 Organic Uremic Solutes

Free Water-Soluble Low-Molecular-Weight Solutes	MW (Da)
Guanidines	
ADMA	202
Argininic acid	175
Creatinine	113
Guanidine	59
Methylguanidine	73
Peptide	
β-Lipotropin	461
Polyols	
Erythritol	122
Myoinositol	180
Sorbitol	182
Threitol	122
Purines	
Cytidine	234
Hypoxanthine	136
Uracil	112
Uric acid	168
Xanthine	152
Pyrimidines	
Orotic acid	174
Thymine	126
Uridine	244
Ribonucleosides	
1-Methyladenosine	281
Pseudouridine	244
Xanthosine	284
Other	
Malondialdehyde	71
Oxalate	90
Urea	60

Protein-Bound Solutes	MW (Da)
AGEs	162
3-Deoxyglucosone	
Fructoselysine	308
Glyoxal	58
Pentosidine	342
Hippurate	
Hippuric acid	179
Indoles	
Indoxyl sulfate	251
Melatonin	126
Quinolinic acid	167
Phenols	
Hydroquinone	110
p-Cresol	108
Phenol	94
Polyamines	
Putrescine	88
Spermidine	145
Spermine	202
Other	
Homocysteine	135

Middle Molecules	MW (Da)
Cytokines	32,000
Interleukin-1β	
Interleukin-6	24,500
Tumor necrosis factorα	26,000
Peptides	
Adrenomedullin	5,729
ANP	3,080
β_2-Microglobulin	11,818
β-Endorphin	3,465
Cholecystokinin	3,866
Cystatin C	13,300
Delta sleep-inducing peptide	848
Hyaluronic acid	25,000
Leptin	16,000
Neuropeptide Y	4,572
PTH	9,225
Retinol-binding protein	21,200
Other	23,750
Complement factor D	

ADMA, Asymmetric dimethylarginine; *AGEs*, advanced glycation end-products; *ANP*, atrial natriuretic peptide; *MW*, molecular weight; *PTH*, parathyroid hormone.

end-products, TNF-α, and phenolic compounds. Clearances of the unbound fraction may be increased by use of HDF, especially postdilutional modes. “Leaky” dialysis membranes and possibly medium cut-off membranes increase the clearance of protein-bound solutes in relationship to the pore size distribution of the membrane.

Middle molecule compounds: Traditionally, middle molecules encompass substances with a molecular weight between 5 and 32 kDa. Their removal is increased when using high-flux dialyzers, high-volume HDF, and medium cut-off membranes.

The clinical benefits of increased removal of protein-bound solutes and specific middle molecules have not been evaluated in appropriate clinical trials.

ADDITIONAL DEVICES AND TECHNOLOGIES

Relative Blood Volume Monitoring

Blood volume monitors provide continuous noninvasive monitoring of relative changes in blood volume by continuously measuring plasma protein concentration using ultrasound or hematocrit by optical scattering. A sharp decline in relative blood volume (RBV) may precede intradialytic hypotension. In the RCT Crit-Line Intradialytic Monitoring Benefit (CLIMB) study, mortality was higher in patients using RBV monitoring compared with conventional monitoring. However, because of an atypically low mortality rate in the conventional monitoring group, the authors pointed out that these findings should be generalized to the HD population in the United States with caution.[6] In an observational study of 842 maintenance hemodialysis patients, specific ranges of RBV ranges have been shown to be associated with improved all-cause mortality.[7] In this study, two-thirds of patients showed a decline of RBV by less than 8% 3 hours into the treatment, possibly indicating fluid overload. In clinical practice, flat RBV curves may prompt a re-evaluation of the target weight. In some dialysis machines, the rate of blood volume change is used to automatically adjust ultrafiltration rates and dialysate sodium concentration (feedback control). This approach has been shown to reduce the frequency of symptomatic intradialytic hypotension in some, but not all, studies.[8,9]

Ultrafiltration Profiling

The ultrafiltration rate is usually kept constant but can be changed during the dialysis session in a preprogrammed manner (ultrafiltration profiling). It may be advantageous to remove a large proportion, such as two-thirds of the prescribed total ultrafiltration volume, in the first half of the HD session. The clinical benefits of ultrafiltration profiling are under debate.

Sodium Profiling

Normally, the dialysate sodium concentration is kept constant throughout the dialysis treatment. The variable sodium option allows dynamic changes of the dialysate sodium concentration during the treatment (sodium profiling). Typically, such treatments would start with a high dialysate sodium concentration (e.g., 145 mmol/L) and end with a lower one (e.g., 135 mmol/L). Some patients with hemodynamic instability may benefit from this option in which the initial sodium concentration is kept high and then slowly reduced to avoid sodium loading. Stepwise profiling was shown to be effective in reducing intradialytic hypotensive episodes.[10] Sodium profiling may result in dialytic sodium loading that may manifest itself in increased thirst, interdialytic weight gain, and blood pressure. In patients presenting with these signs and symptoms, the need for sodium profiling should be critically re-evaluated. Using conductivity measurements of dialysate, some modern dialysis machines allow the user to target a specific dialytic sodium balance.

Online Clearance Monitoring

Sodium and urea clearances are identical for practical purposes. The conductivity of the dialysate is largely a function of the dialysate sodium concentration, and online clearance monitors use this feature to compute the urea clearance (K) of a dialyzer during a dialysis treatment.

Blood Temperature Monitoring and Dialysate Cooling

In most patients, HD exerts a net positive thermal balance that may affect hemodynamic stability. “Cool” dialysate has been shown to improve vascular stability during dialysis. An RCT showed cardiac and brain white matter protection with individualized cooling at 0.5°C below body temperature.[11,12] The most precise control of a patient’s temperature during dialysis is enabled by a body temperature monitor (BTM) that adjusts in a feedback loop the dialysate temperature to attain a prescribed thermal balance and core temperature. Alternatively, tympanic temperature can be used to prescribe the dialysate temperature (e.g., 0.5°C below tympanic temperature). In some studies, the dialysate temperature was set to 35°C irrespective of the patient’s body temperature.

Intradialytic Oxygen Measurement

Blood coming from a functioning arteriovenous access resembles arterial blood. Some devices and dialysis machines measure predialyzer blood oxygen saturation. In an observational study, patients with prolonged intradialytic hypoxemia showed greater morbidity and mortality.[13] In HD patients with central venous catheter as vascular access, a low central venous oxygen saturation is associated with intradialytic hypotension.[14]

HOME HEMODIALYSIS

Home HD (HHD) offers several clinical benefits compared with conventional renal replacement therapies, such as better survival, enhanced blood pressure control and left ventricular geometry, normalization of phosphate balance, augmentation of kidney-specific quality of life scores, and improved fertility.[15] HHD also provides greater patient autonomy in a cost-effective manner.[16] There are variations in HHD prescription and practices (Table 98.6). Additionally, conventional HD platforms and low dialysate flow systems may be used in the home.

Several considerations should be addressed before starting patients on HHD. Routinely, a home visit should be conducted before

TABLE 98.6 Home Hemodialysis Prescription and Practices

	Conventional	Short Daily	Nocturnal	Low Dialysate Flow Systems
Treatments per week	3	6	5–6	6
Treatment time (hours)	4	2–3	6–8	2.5–3.5
Blood flow rate (mL/min)	400	400	200	400
Dialysate flow rate (mL/min)	500	800	300	130

starting home dialysis to assess for feasibility and the potential need for renovations and modifications. The Advancing American Kidney Health Initiative was announced in 2019, which aimed to ensure that 80% of incident patients with end-stage kidney disease will receive home-based dialysis or a kidney transplant by 2025. Given the change in policy, it is anticipated that there will be a global growth in home therapies.

DIALYSIS MACHINE CHOICE AND OTHER EQUIPMENT

The choice of machine should be tailored to the patient's individual requirements and preference, and it will also determine the ease of dialysis fluid preparation. Additionally, the availability of appropriate water, electrical supply, and space may influence the choice of HD machine. Conventional HD platforms will require water filtration systems, whereas others (i.e., low-dialysate flow systems) may use prepackaged dialysate or generate online dialysis solutions. Premixed bags are available with bicarbonate-buffered or lactate-buffered dialysate. All types of vascular accesses may be used for HHD. Registry data suggest that permanent vascular access has superior outcomes compared with tunneled HD catheters. Moreover, the use of buttonhole cannulation (compared with the stepladder technique) is associated with higher infectious complications and hence should be reserved for selected patients only.

WATER PREPARATION, STANDARDS, AND PLUMBING

As for in-center HD, prevailing standards for water quality should be adhered to (see previous discussion). Depending on the local water conditions and regulations, water softener, backflow preventers, and blending devices may be necessary in the home. In certain circumstances, feeder tanks may be necessary to provide the water pressure necessary for the RO unit and dialysis machine to function properly.

SAFETY

All dialysis machines are equipped with monitors, as described earlier. Leak and moisture detectors should be placed around vascular access cannulation sites, under the dialysis machine, and at the water treatment system. For central venous catheters, special connectors or catheter safety lock boxes can be used. Telemonitoring systems have been employed for HHD, but they are not a prerequisite. Practically, most HHD users continue to rely on an "on-call" system for troubleshooting. Typically, arterial and venous pressure alarms are the most common type of alarms. The average number of alarms per night decreased significantly over time as patients gained experience with HHD (from a maximum of 1.98 ± 3.31 alarms/night to a low of 0.74 ± 1.63 alarms/night) according to the London Ontario experience.

BOX 98.1 Activities to Reduce the Environmental Footprint of Hemodialysis

Electricity
- Reporting of energy consumption
- Energy-efficient dialysis machine implementation
- Use of light sensors and lighting timers
- Switch to LED bulbs
- Reduced facility size
- Integration of dialysis unit into a high environmental-quality building
- Change or tuning of air treatment systems

Water
- Reporting of water usage
- Water-efficient dialysis machine implementation
- Change in water treatment system

Care-Related Waste
- Reporting of waste produced
- Regular audits of waste management
- Caregiver training for waste sorting

LED, Light-emitting diode.
Modified from Bendine G, Autin F, Fabre B, et al. Haemodialysis therapy and sustainable growth: a corporate experience in France. *Nephrol Dial Transplant.* 2020;35:2154–2160.

WEARABLE ARTIFICIAL KIDNEY

Wearable artificial kidneys (WAKs) have been successfully tested in clinical trials, proving that this technology is safe and a potential future form of renal replacement therapy. At their current stage, however, WAKs are not yet suitable for routine use, facing formidable challenges such as anticoagulation, toxin clearance, and fluid removal.

"GREEN" HEMODIALYSIS

Hemodialysis is a resource-intensive treatment that uses substantial amounts of energy and water and produces plastic and other waste. The annual per-patient carbon footprint of 6-chair satellite dialysis unit has been calculated to be 10.2 tonnes of CO_2 equivalents. Importantly, environmental responsibility awareness, staff education, and improved technology can substantially improve the environmental footprint of hemodialysis. By implementing systemwide changes (Box 98.1), between 2005 and 2018, a French dialysis network was able to reduce water use from 801 to 382 L/session, power use from 23.1 to 16.3 kWh/session, and waste from 1.8 to 1.1 kg/session.[17] Additional measures call for recycling reverse osmosis water, renewable energy use, commingled recycling, polyvinyl chloride plastic recycling, secured bicycle parking, shower and changing facilities for bike riders, videoconferencing, and telemedicine. Little is known about the ecological impact of peritoneal dialysis. "Green" dialysis is likely to evolve as a major topic in the years to come.

SELF-ASSESSMENT QUESTIONS

1. Which statement is *correct?*
   A. Bacterial endotoxin in the HD fluid cannot pass through the dialysis membrane.
   B. An increase in HD fluid sodium concentration will always raise the serum sodium level.
   C. HD fluid buffers include anions such as bicarbonate diacetate, chloride, and citrate.
   D. HD fluid potassium concentrations of 1 mEq/L are safer than 3 mEq/L.
2. Which statement is *correct?*
   A. A high-protein diet will eventually reduce the level of uremic toxins in the blood.
   B. Urea is considered a powerful uremic toxin.

C. Some uremic toxins are synthesized in the colon and absorbed into the bloodstream.
D. HDF is highly efficient in removing protein-bound uremic toxins.

3. Which statement is *correct?*
   A. A typical symptom of hard water syndrome is spiking fever during hemodialysis.
   B. Medium cut-off membranes showed significantly improved survival in a randomized controlled trial.
   C. Predilution HDF is most efficient in terms of increasing solute removal.
   D. Dialysate cooling is an effective means to increase hemodynamic stability.
4. Which statement is *correct?*
   A. Intradialytic hypoxemia is associated with poor outcomes.
   B. Sodium profiling is a preferred way to treat patients with intradialytic hypertension.
   C. In patients with predialysis serum potassium levels greater than 6.2 mEq/L, a dialysate potassium concentration of 0 mEq/L is warranted.
   D. Home HD is not advised in patients with diabetes mellitus.
5. Which statement is *correct?*
   A. A positive venous pressure is a reliable indicator of correct needle placement.
   B. Water treatment is key to the preparation of adequate dialysis fluid.
   C. Intradialytic hypoxemia is associated with poor outcomes.
   D. Modern dialyzer membranes combine polysulfone with cellulose.

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